INCH-POUND
MIL-PRF-3098/78F
7 January 2010
SUPERSEDING
MIL-PRF-3098/78E
12 January 2005

PERFORMANCE SPECIFICATION SHEET

# CRYSTAL UNIT, QUARTZ, CR6/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 2 MHz to 10 MHz; fundamental; noncontrolled; antiresonance.

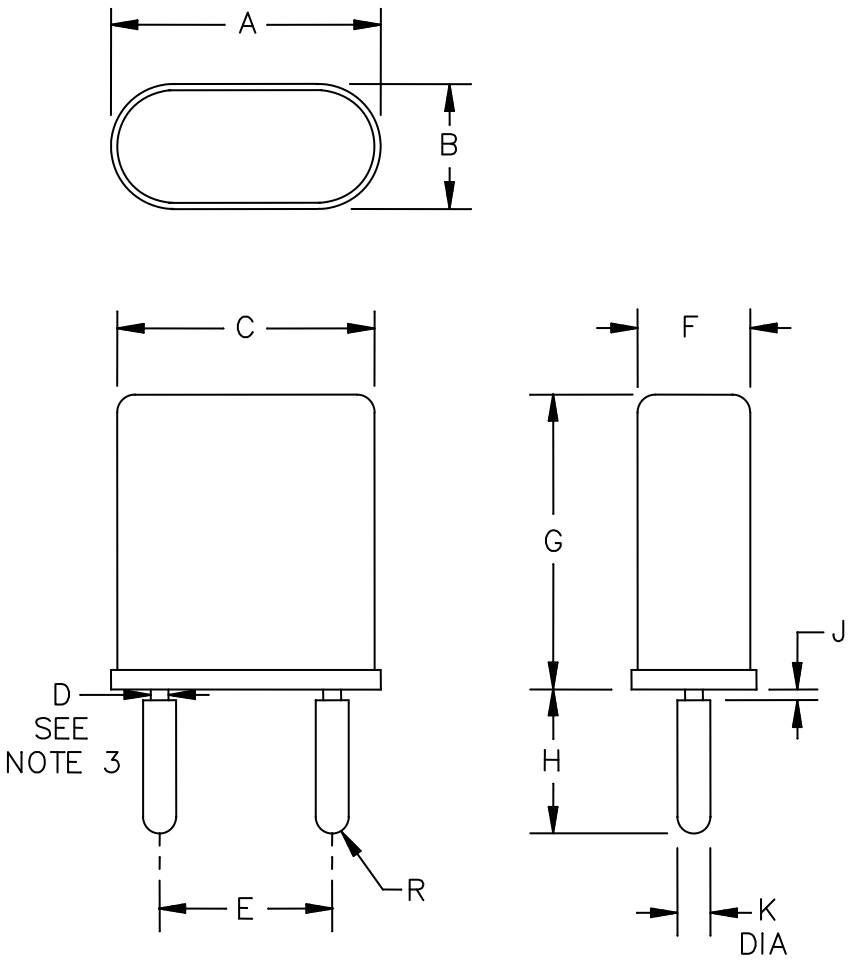


FIGURE 1. Crystal unit - CR6/U.



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| Ltr | Inches | | mm | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | --- | .757 | --- | 19.23 |
| B | --- | .352 | --- | 8.94 |
| C | --- | .752 | --- | 19.10 |
| D | .048 | .052 | 1.22 | 1.32 |
| E | .478 | .494 | 12.14 | 12.55 |
| F | --- | .317 | --- | 8.05 |
| G | --- | .775 | --- | 19.68 |
| H | --- | .448 | --- | 11.38 |
| J | .012 | .047 | 0.30 | 1.19 |
| K | .091 | .095 | 2.31 | 2.41 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR6/U - Continued.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.0 MHz to 10.0 MHz, inclusive.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW maximum.

Operating temperature range (noncontrolled): -40°C to +70°C, inclusive.

Frequency tolerance: ±50 parts per million (ppm).

Equivalent resistance: See table I.

Antiresonance, load capacitance: 13.0 pF ±0.5 pF.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.



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TABLE I. Equivalent resistance.

| Frequency range, Inclusive | Maximum resistance |
|---|---|
| MHz | Ohms |
| 2.00 to 2.1 | 940 |
| 2.1+ to 2.2 | 860 |
| 2.2+ to 2.25 | 820 |
| 2.25+ to 2.3 | 780 |
| 2.3+ to 2.4 | 720 |
| 2.4+ to 2.5 | 660 |
| 2.5+ to 2.6 | 600 |
| 2.6+ to 2.7 | 560 |
| 2.7+ to 2.8 | 520 |
| 2.8+ to 3.0 | 450 |
| 3.0+ to 3.2 | 400 |
| 3.2+ to 3.4 | 360 |
| 3.4+ to 3.7 | 300 |
| 3.7+ to 4.0 | 250 |
| 4.0+ to 4.5 | 200 |
| 4.5+ to5.1 | 160 |
| 5.1+ to 6.2 | 110 |
| 6.2+ to 6.7 | 90 |
| 6.7+ to 7.2 | 80 |
| 7.2+ to 7.5 | 70 |
| 7.5+ to 8.0 | 65 |
| 8.0+ to 9.0 | 50 |
| 9.0+ to 10.0 | 45 |

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Part or Identifying Number (PIN): CR6/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202



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The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC, SH
Air Force - 19, 84

Preparing activity:
Army - CR

Agent:
DLA - CC

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.



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